



NANOHUNTER II

Total reflection X-ray fluorescence spectrometer

Benchtop TXRF for ultra-trace elemental analysis



Rigaku

POWERING NEW PERSPECTIVES

NANOHUNTER II – Advanced benchtop TXRF / GI-XRF

Rigaku Corporation, the world's leading X-ray analytical instrumentation company, introduced our first total reflection X-ray fluorescence (TXRF) spectrometer, System 3271, in 1988 for the characterization of surface contamination on semiconductor wafers. Today Rigaku is the leading supplier of TXRF metrology instrumentation to the global semiconductor industry.

First benchtop TXRF/GI-XRF for ultra-trace elemental analysis

Winner of the prestigious 2007 R&D 100 award, the original Rigaku NANOHUNTER benchtop spectrometer offered comprehensive trace element and materials characterization analysis capabilities to a broader range of research disciplines, and in more diverse analytical settings, than was possible with previous technology.

Next generation NANOHUNTER II

Now the completely redesigned, next generation NANOHUNTER® II spectrometer offers over an order of magnitude (12X) greater X-ray tube power, together with a high precision goniometer, to enable unparalleled high-sensitivity ultra-trace elemental analysis in both TXRF and grazing incidence X-ray fluorescence (GI-XRF) modes.

Whether for geologists, chemists, biochemists, biologists, materials scientists or engineers, non-destructive trace element analysis is attainable, with minimal to no sample preparation, for applications that span from metallo-protein research to environmental assessment and thin-film metrology.

Ultra-trace elemental analysis

The NANOHUNTER II spectrometer combines a fully automatic optical axis adjustment system that provides stable high-sensitivity analysis in an easily handled benchtop form factor that allows quick and trouble-free operation. With a high-power 600 W X-ray source, a newly developed mirror (optic) and a large-area silicon drift detector (SDD), the NANOHUNTER II spectrometer delivers high signal-to-noise measurements. The lower limit of Cd detection is 8 ppb ($\mu\text{g/L}$). For As and Se in a liquid, a lower detection limit of 0.8 ppb or less has been achieved.

Unique Rigaku optic delivers dual energy excitation

One important advantage is the high efficiency with which $K\alpha$ lines can be excited. Until now, efficiently exciting $K\alpha$ lines in ^{48}Cd has been difficult, so TXRF users relied on analysis using L-lines, which overlap with other commonly occurring elements. A new multilayer monochromator has been specially designed to reflect X-rays with two energies: the Mo $K\alpha$ line (17.4 keV) and higher energy bandpass of about 30 keV. The higher energy X-rays can effectively excite K line fluorescent X-rays of heavy elements from Ru to Sn. Wide adoption of this capability is expected in fields such as the screening and analysis of factory waste liquids and the analysis of beverages, as such as wine.

Easy non-technical operation for routine aqueous analysis

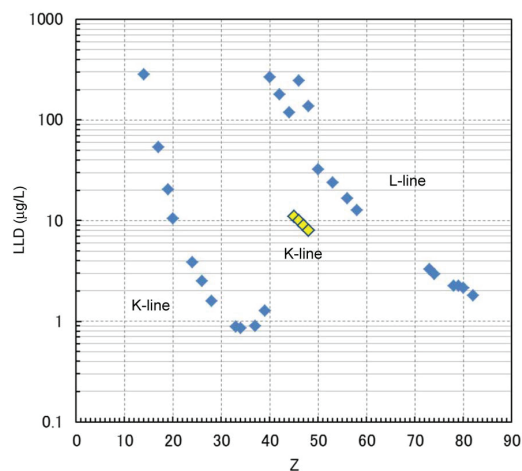
The most common application of TXRF is elemental analysis of aqueous samples using the "Drop & Dry" sample preparation method shown at right: A) obtain aliquot of sample, B) add known amount of an internal standard, C) mix, D) remove $\approx 10 \mu\text{L}$ sample, E) pipette onto a clean carrier, F) dry and G) measure.^[1]



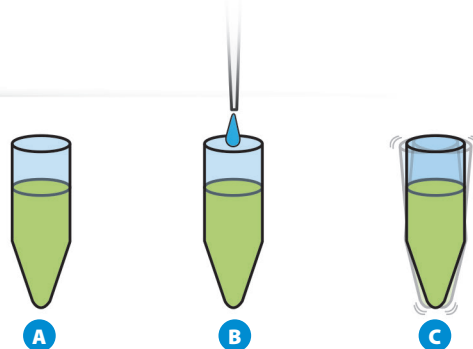
Rigaku NANOHUNTER II with variable incident angle optics and a 600 W X-ray tube source



"Mainframe" Rigaku TXRF-310 for new semiconductor wafers



Lower limit of detection (LLD) as a function of atomic number (Z)



What is X-ray fluorescence

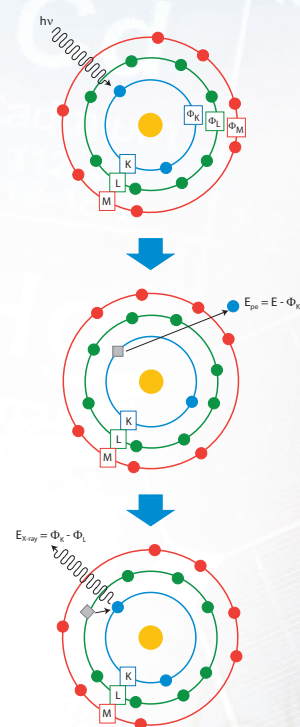
In X-ray fluorescence (XRF), an electron can be ejected from its atomic orbital by the absorption of X-ray (photon) from an X-ray tube. The energy of the photon ($h\nu$) must be greater than the energy with which the electron is bound to the nucleus of that atom. When an inner orbital electron is ejected from an atom (middle image), an electron from a higher energy level orbital transfers to fill the vacant orbital. During this transition, a photon may be emitted (right image).

Because the energy difference between two specific orbital shells is always the same for a specific element, the emitted photon will always have the same characteristic energy (keV). For a fluorescent emission line, for a given element, the number of photons per unit time (counts per second or cps) is related to the amount of that element in a sample. Counting rates are calculated by measuring, for a set time, the number of photons detected for the various observed elemental X-ray fluorescence lines (spectral peaks). Thus, qualitative and quantitative elemental analysis is achieved by determining the energy of X-ray peaks in a sample spectrum and measuring their associated count rates.

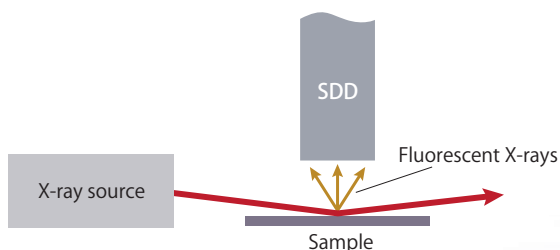
How TXRF works

An incident X-ray beam impinges upon the sample at a shallow angle (below the critical angle for total reflection of X-rays for the substrate, $\approx 0.1^\circ$) resulting in virtually complete reflection of the excitation beam away from the silicon drift detector (SDD).

This affords dramatically reduced background contributions in the measured energy dispersive X-ray fluorescence (EDXRF) spectra. By efficiently exciting the surface elements, while virtually eliminating the background noise, the TXRF technique affords extremely high signal-to-noise performance, resulting in ultra-trace elemental measurement sensitivity.



Theory of XRF spectrometry



Schematic of the TXRF experiment



D



E



F



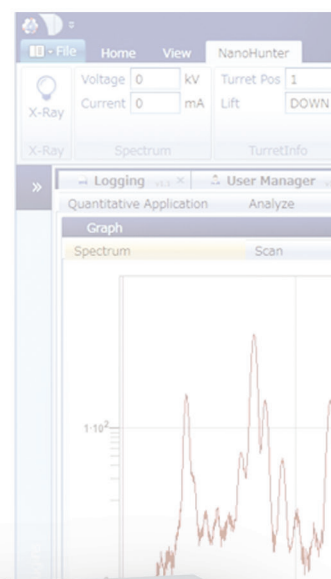
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Benchtop TXRF / GI-XRF for ultra-trace elemental analysis

NANOHUNTER II

Aluminum	Silicon	Phosphorus	Sulfur	Chlorine	Argon
Al 13	Si 14	P 15	S 16	Cl 17	Ar 18
Atomic Weight: 26.981 538 6	Atomic Weight: 28.085 579 6	Atomic Weight: 30.973 761 99	Atomic Weight: 32.06 5	Atomic Weight: 35.453	Atomic Weight: 39.948 163 4
Gallium	Germanium	Arsenic	Selenium	Bromine	Krypton
Ga 31	Ge 32	As 33	Se 34	Br 35	Kr 36
Atomic Weight: 69.723 1	Atomic Weight: 72.640 08	Atomic Weight: 74.921 601 6	Atomic Weight: 78.96 18	Atomic Weight: 79.904 1	Atomic Weight: 83.801 881 1
Rubidium	Sr	Yttrium	Zirconium	Niobium	Molybdenum
Rb 37	Sr 38	Y 39	Zr 40	Nb 41	Mo 42
Atomic Weight: 85.467 8	Atomic Weight: 87.62	Atomic Weight: 88.905 84	Atomic Weight: 91.224 2	Atomic Weight: 92.906 38	Atomic Weight: 95.94
Cesium	Barium	Lanthanum	Cerium	Praseodymium	Neodymium
Cs 55	Ba 56	La 57	Ce 58	Pr 59	Nd 60
Atomic Weight: 132.905 451 9	Atomic Weight: 137.327	Atomic Weight: 138.904 71	Atomic Weight: 140.127 08	Atomic Weight: 140.907 65	Atomic Weight: 144.242 0
Francium	Radium	Actinium	Thorium	Protactinium	Uranium
Fr 87	Ra 88	Ac 89	Th 90	Pa 91	U 92
Atomic Weight: 223.018 51	Atomic Weight: 226.025 4	Atomic Weight: 227.027 71	Atomic Weight: 232.037 74	Atomic Weight: 231.036 888 2	Atomic Weight: 238.028 91



Analyze Al through U

Broad elemental analysis coverage for maximum analytical flexibility.

Low part-per-billion detection limits

^{48}Cd LLD is 8 ppb. For ^{33}As and ^{34}Se in liquids, the LLD is 0.8 ppb or less.

16-position sample changer

Accepts standard 30 mm diameter sample carriers as well as standard microscope slides.

Digital data output

Data export and LIMS compatibility are supported using either RS-232C or TCP/IP.

Large area silicon drift detector

High-resolution and large area thermo-electrically cooled SDD delivers superior counting statistics for improved detection limits.

64-bit Windows® based software

Rigaku SmartLab Studio II is the new 64-bit software suite for measurement and analysis of data obtained from different X-ray analytical techniques. Equipped with the NANOHUNTER plugin, it is both extraordinarily powerful and extremely easy to use. The depth and breadth of features, as well as the sophistication of the interface, is the result of decades of XRF software development at Rigaku.





Applications

TXRF is a well-established analytical technique for multi-element determination in various types of samples, especially liquids, powders and micro samples. To perform analysis under total-reflection conditions, samples must be provided as thin films. For liquid samples, this is done by depositing 5–50 μL of sample on a reflective carrier and then drying the drop prior to measurement. For non-aqueous samples, the scientific literature is replete with TXRF sample preparation protocols.

Applications cover a broad range of sample types spanning from traditional environmental TXRF uses to food, biology, medical, pharmaceutical, industrial, archaeometry and forensic science. Examples of the utility of TXRF include elemental fingerprint authentication in fields such as pharmaceuticals, fine chemicals and forensics. Detection of heavy metal contamination in food, beverages and food ingredients comprise yet another set of applications. Integrity of nutraceuticals, vitamins, “traditional” medicines and dietary supplements are important to customs laboratories, importers, formulators and resellers. Contamination analysis of wafers, solar cells, OLEDs and nano-particles encompass the electronics and materials science fields. Quality control of pure substances and industrial products as well as tissues and biological liquids in the clinical environment are also important.

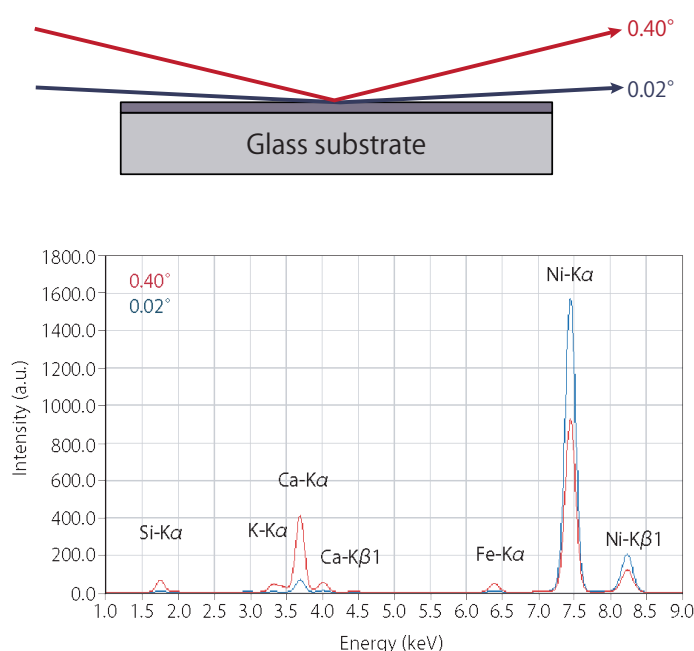
And, of course, environmental analysis continues to be a main driver of TXRF adoption. From contamination of water and soil through R&D and quality control, this unique technology provides unique insights with low costs of operation.

Applications ^[2]	Sample types
Environmental and geological	Airborn particulates
	Biomonitoring
	Sediment, soil, sludge, ores, rock, minerals
	Water, rain, wastewater, effluent
Food and biological	Beverages: water, tea, coffee, beer, wine, juice
	Plant materials and extracts
	Unicellular organisms and biomolecules
	Vegetable and essential oils
Medical and pharmaceutical	Blood, plasma, serum, etc.
	Medicines, IV fluids and OTC pharmaceuticals
	Packaging materials used to increase shelf life
	Urine, cerebrospinal and amniotic fluid
Industrial and chemical	Antimicrobial and antifungal surfaces
	Ceramic powders
	Hydrocarbons
	Nanostructured surfaces
	Solvents, acids, bases
	Ultra-pure reagents
	Wafer surfaces
Archaeometry and forensics	Drugs and poisons
	Metals and alloys
	Pigments, inks, and varnishes
	Textile fibers and glass splinters
	Traditional medicines
	Unknown materials

Variable angle GI-XRF capability

In the area of analysis of solid surfaces, there is demand (primarily in the fields of thin films and thick films) for analyses that penetrate slightly deeper than the surface. For these kinds of analyses, a methodology called grazing incidence X-ray fluorescence (GI-XRF) is employed, whereby the elements beneath the surface are excited by varying the incident angle of the X-ray source. Because the NANO HUNTER II TXRF spectrometer has variable angle of incidence, it is possible to perform depth profile surface analyses. The GI-XRF technique is applicable to nanoscale research, as well as film and coatings metrology.

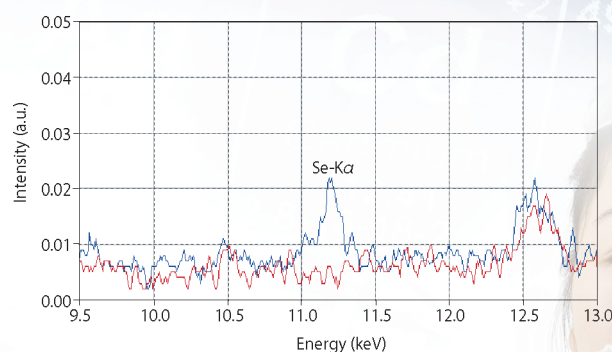
In this example, a 25 nm nickel (Ni) thin film on a glass substrate was measured at two different grazing angles: 0.02° and 0.40°. Only Ni was strongly excited at 0.02° GI excitation condition (blue spectrum). However, elements included in the glass substrate are excited at the higher 0.40° angle (red spectrum) because X-rays penetrate more deeply into the substrate.



Examples

Environmental analysis (effluent)

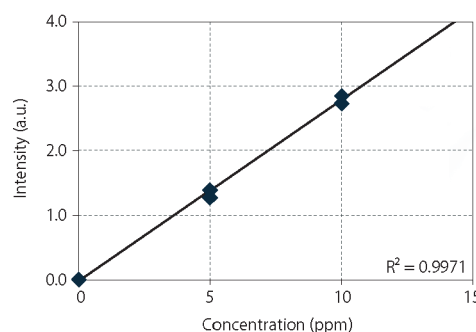
According to standards for hazardous pollutants in effluent, set by the Ministry of the Environment in Japan, arsenic (As) and selenium (Se) limits are set below 0.1 mg/L. The spectrum of 10 ppb Se in water solution clearly shows a prominent $K\alpha$ peak relative to the background (red spectrum). From this measurement, the calculated lower limit of detection (LLD) is about 0.8 ppb. The sample was easily prepared by pipetting 10 μ L of aqueous solution onto a glass substrate followed by drying. Large numbers of samples may be quickly prepared and measured with this "Drop & Dry" sample preparation technique.



Selenium in effluent stream

Analysis of impurities in pharmaceuticals

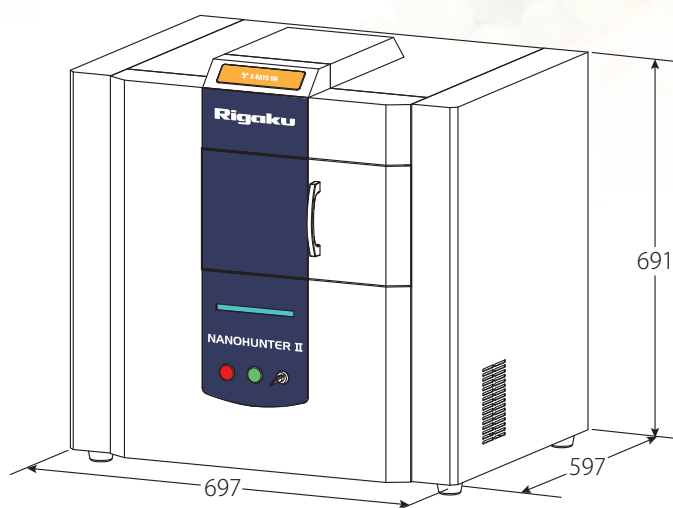
The maximum allowable concentration of inorganic elemental impurities in pharmaceuticals is severely limited by regulations like USP <232>. In this example, analysis of palladium (Pd) in solution is illustrated because residual Pd catalyst can be a common impurity. With a traditional direct excitation XRF spectrometer, it is difficult to obtain high signal-to-noise spectra for elements like Pd or rhodium (Rh) due to high background intensities in the K-line energy range (20 - 22 keV). However, with the TXRF technique, it is possible to dramatically reduce the background present in the measured spectra. In this example, the calibration curve was calculated from measurements on samples made by pipetting 10 μ L of 5 ppm and 10 ppm respective aqueous Pd standards onto glass substrates followed by drying. A lower limit of detection (LLD) of 2 ppb was obtained from this curve.



Analysis of palladium in solution

Specifications

NANO HUNTER II		
X-ray tube	Rated output	600 W
	Target material	Mo
XG	Voltage / Current	50 kV - 12 mA
Optics	Monochromator	Double stacked
	Grazing angle	-0.05° - +0.45°
Detector	Type	SDD (Peltier thermo-electric cooling)
Sample chamber	Atmosphere	Air, N ₂ *, He*
	Turret	16-position
	Sample size	26 x 76 mm, Ø30 mm
	Thickness	Max. 5 mm
	Measuring area	Ø10 mm
Cooling water	1 kW cooling water external circulator	
OS / Computer	Windows® / PC, Monitor	
Power	100 - 240 VAC ±10%, 1-phase, 50/60 Hz	
Dimensions	697 W x 691 H x 597 D (mm)	
Weight	Approx. 90 kg	



Backed by Rigaku

Since its inception in 1951, Rigaku has been at the forefront of analytical and industrial instrumentation technology. Today, with hundreds of major innovations to our credit, the Rigaku Group of Companies are world leaders in the field of analytical X-ray instrumentation. Rigaku employs over 1,400 people worldwide in operations based in Japan, the U.S., Europe, South America and China.

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Total reflection X-ray fluorescence spectrometer



www.rigaku.com/products/txrf



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References

¹ For a more detailed discussion of the Drop & Dry technique, see page 247 in *Total-Reflection X-ray Fluorescence Analysis and Related Methods*, Second Edition, by Reinhold Klockenkämper and Alex von Bohlen (Published by John Wiley & Sons, Inc.).

² Klockenkämper, Reinhold, and Alex von Bohlen. *Total-Reflection X-ray Fluorescence Analysis and Related Methods*, Second Edition. John Wiley & Sons, Inc., 2015. p. 291.